

Apache Kafka Interview Questions and Answers

Interview questions covering Kafka fundamentals: topics, partitions, brokers, consumer groups, offsets, replication, and related tooling such as Kafka Connect and Kafka Streams.

Q1. What is Apache Kafka?

Apache Kafka is a distributed event streaming platform used to publish, store, and process streams of records in a fault-tolerant and scalable way. It is commonly used for building real-time data pipelines and event-driven applications.

Q2. What is a topic and a partition in Kafka?

A topic is a named category to which records are published. Each topic is split into one or more partitions, which are ordered, append-only logs. Partitioning allows a topic's data to be distributed and processed in parallel across multiple brokers and consumers.

Q3. What is the role of a broker?

A broker is a Kafka server that stores data and serves client requests. A Kafka cluster typically consists of multiple brokers, and partitions are distributed across them to balance load and provide fault tolerance through replication.

Q4. What is a consumer group?

A consumer group is a set of consumers that cooperate to read data from a topic, with each partition consumed by only one member of the group at a time. This allows horizontal scaling of consumption while ensuring each record is processed once within the group.

Q5. How does Kafka track a consumer's position in a partition?

Kafka tracks progress using an offset, a sequential id assigned to each record within a partition. Consumers commit offsets (to an internal Kafka topic) to record how far they have read, so they can resume from the correct position after a restart.

Q6. What is the replication factor?

The replication factor determines how many copies of each partition are maintained across different brokers. A higher replication factor improves fault tolerance, since the partition remains available if some replicas' brokers fail, at the cost of additional storage and network overhead.

Q7. What is the difference between ZooKeeper mode and KRaft mode?

Older Kafka versions relied on Apache ZooKeeper to store cluster metadata and coordinate brokers. KRaft (Kafka Raft) is a newer mode that removes the ZooKeeper dependency by managing metadata internally using the Raft consensus protocol, simplifying deployment and operations.

Q8. What is Kafka Connect?

Kafka Connect is a framework for reliably streaming data between Kafka and external systems, such as databases, file systems, or search indexes, using reusable source and sink connectors instead of writing custom producer or consumer code.

Q9. What is Kafka Streams?

Kafka Streams is a client library for building stream-processing applications directly on top of Kafka. It supports operations like filtering, aggregating, and joining streams, and treats state and processing as part of

the same fault-tolerant, distributed model as Kafka itself.

Q10. How does Kafka provide message ordering guarantees?

Kafka guarantees ordering only within a single partition, not across an entire topic. Records with the same key are always sent to the same partition, so ordering is preserved for records sharing a key, while records across different partitions may be read in a different relative order.